

GroMo25: ACM Multimedia 2025 Grand Challenge for Plant Growth Modeling with Multiview Images

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Abstract

Understanding plant growth dynamics is a critical component of modern agricultural research, with applications in yield prediction, phenotyping, and sustainable crop management. Despite recent advances in computer vision and deep learning, progress in plant growth modeling has been constrained by the lack of publicly available, high-resolution, multiview, and temporally rich datasets. To address this gap, we introduce **GroMo25**, the first international challenge on *plant growth modeling using multiview imagery*. In this challenge, we propose a dataset that comprises images of four crops: wheat, mustard, radish, and okra, captured at consistent time intervals from multiple camera viewpoints under controlled environmental conditions. The challenge focuses on two key tasks: (1) *plant age prediction* and (2) *leaf count estimation*, both requiring models to use spatial and temporal plant features. GroMo25 attracted participation from multiple teams worldwide, encouraging benchmarking and innovation in vision-based plant phenotyping. The GitHub repository is publicly available at <https://github.com/mriglab/GroMo-Plant-Growth-Modeling-with-Multiview-Images>.

CCS Concepts

• Computing methodologies → Artificial intelligence .

Keywords

Growth age prediction, Leaf count estimation, Multiview

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1 Introduction

Plant growth monitoring is a critical component of modern plant breeding [7], precision agriculture [3], and yield estimation [18]. Accurate and consistent assessment of plant development empowers researchers and agronomists to make informed decisions regarding plant health, stress responses, and productivity. Growth can be inferred through morphological changes in specific plant organs, such as leaves [6], flowers [8], stems [10], and fruits [9], which serve as visual cues of plant development. Among these, leaf counting has emerged as a fundamental and widely adopted metric in high-throughput plant phenotyping [13]. The number and shape of leaves provide direct clues about the growth stage and physiological status of a plant. Consequently, automating the leaf count process through computer vision and deep learning techniques has gained considerable attention, particularly in controlled growth environments where thousands of plants must be efficiently analyzed.

Several deep learning-based approaches have been developed to address the challenge of leaf counting, utilizing a variety of architectures such as convolutional neural networks (CNNs) [13, 14], encoder-decoder models, and object detection frameworks [1, 2, 5, 11, 16, 17]. These methods range from direct regression techniques to multi-stage pipelines that incorporate segmentation followed by counting. Lightweight models like Tiny-YOLOv3 have been used for efficient localization [5], while more complex networks such as Eff-UNet++ [2] and two-stream pyramid networks [16] offer improved



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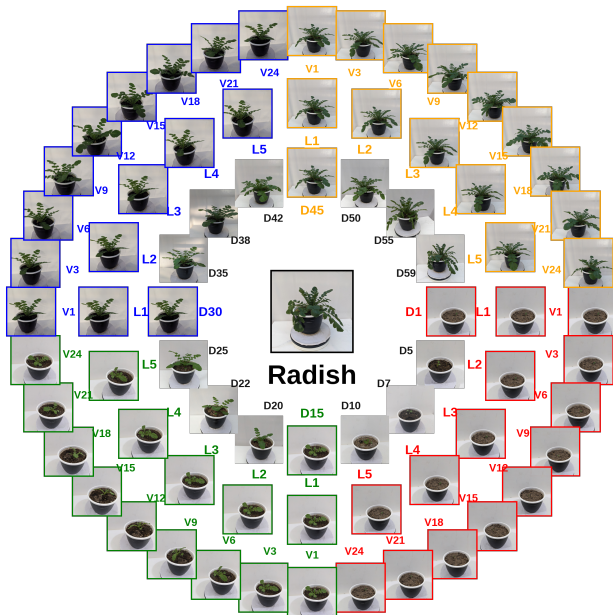


Figure 1: Visualization of the GroMo'25 dataset. The central image represents the radish plant, surrounded by three concentric rings. The first ring (Circle 1) shows temporal snapshots (D1–D59) across 16 representative days. The second ring (Circle 2) groups selected days into five camera height levels (L1–L5). The outermost ring (Circle 3) depicts multiview observations (V1–V24) from nine representative viewpoints across four (red (D1), green (D15), blue (D30), and yellow (D45)) key time points.

performance by combining spatial detail and hierarchical feature extraction. Many of these approaches have been evaluated on publicly available datasets, including Arabidopsis [5], KOMATSUNA [20], MSU-PID [12], and CVPPP [19], which provide diverse plant structures and imaging conditions to test model generalizability. Despite their performance, a critical limitation persists across these models: the exclusive reliance on single-view images, typically captured from a fixed overhead angle. This restricts the model's ability to handle occlusions and impairs structural comprehension in scenarios involving dense or overlapping canopy regions. Moreover, existing publicly available datasets often lack temporal sequences, thereby failing to capture the progression of growth over time, a crucial aspect of developmental phenotyping. In particular, there is no established benchmark currently for estimating the age of a plant purely from its visual appearance, an overlooked but essential component for time-aware phenotyping and growth forecasting.

In GroMo25: Plant Growth Modeling with Multiview Images, we address these challenges by introducing a new dataset that combines multiview and temporal imagery for comprehensive plant phenotyping. Specifically, we propose two novel tasks:

- **Leaf Counting from Multi-View Images:** Using spatial diversity in views to alleviate occlusion and improve the robustness of the estimation.

Table 1: Summary of models, datasets, and challenges for plant phenotyping tasks.

Model Used	Dataset	Challenge
Fusing Network Components [13]	Dobrescu et al. 2017	Counting Challenge (LCC)
VGG-16 network [14]	CVPPP 2017 dataset (Plant Phenotyping)	CVPPP Leaf Counting Challenge
Tiny-YOLOv3 network [21]	Arabidopsis thaliana	-
Deconvolutional network [1]	CVPPP-2017 dataset	CVPPP Leaf Counting Challenge
Eff-UNet++ [2]	KOMATSUNA, MSU-PID, CVPPP-2017	CVPPP Leaf Counting Challenge
Two-stream CNN [16]	CVPPP 2017 dataset	CVPPP Leaf Counting Challenge
LC-Net [11]	CVPPP, KOMATSUNA datasets	CVPPP Leaf Counting Challenge

- **Plant Age Estimation from Visual Data:** Predicting the chronological age (in days) of a plant using visual cues across multiple views and time points.

Our dataset consists of plants imaged from 24 distinct viewpoints per time point, collected across several days, enabling both spatial and temporal reasoning. This configuration supports spatiotemporal modeling, multiview fusion, and sequential learning. Figure 1 shows the structure of the dataset, illustrating concentric rings of daily captures and viewpoint distributions. A quantitative summary of the dataset, including the number of days, views, plant instances, and labeled traits, is provided in Table 2. We believe that GroMo25 introduces a unique benchmark for the community, encouraging innovation in visual modeling of plant growth. By addressing gaps in existing datasets namely, lack of multiview data, temporal sequences, and plant age annotations this challenge opens new research directions in multimedia-based plant phenotyping and developmental modeling.

2 Challenge Description

This section provides an overview of the GroMo25 dataset and the associated tasks designed to advance plant growth modeling using multi-view temporal imagery, followed by data partitioning and evaluation metrics.

Table 2: Dataset Summary

Crop	Plants	Max Days	Levels	Angles
Wheat	4	118	5	0°-360°(step 15°)
Mustard	4	50	5	0°-360°(step 15°)
Radish	5	59	5	0°-360°(step 15°)
Okra	2	86	5	0°-360°(step 15°)

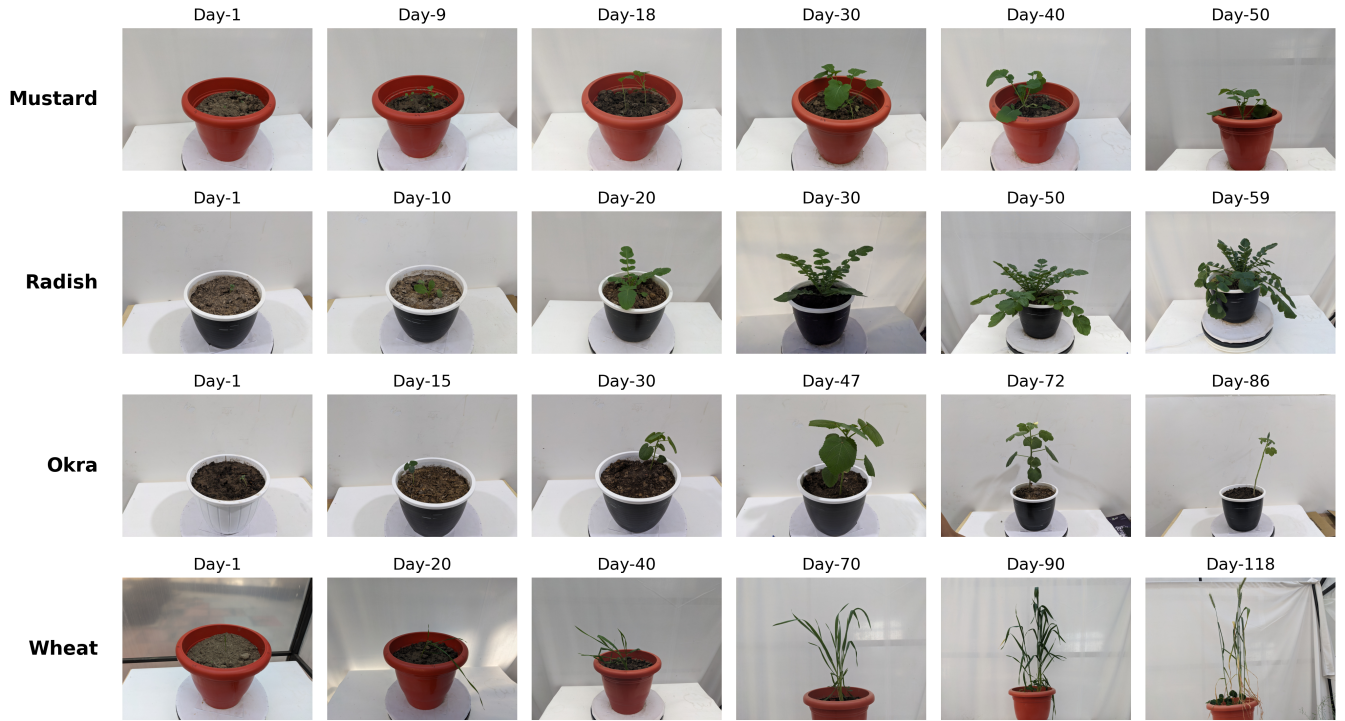


Figure 2: Visual growth progression of individual plants from four crop species — Mustard, Radish, Okra, and Wheat — over time. Each row corresponds to a different crop, and each column shows a representative image of the same plant taken at increasing days after sowing, demonstrating phenotypic changes during development.

2.1 Dataset

The GroMo25 dataset is a high-resolution ($4032 \times 3024 \times 3$), multi-view image dataset capturing the growth lifecycle of four diverse crops: wheat, mustard, radish, and okra. The dataset is collected in a controlled environment, addressing challenges in plant phenotyping, growth modeling, and agricultural AI. It offers a rich set of plant images taken at consistent intervals, angles, and heights. This dataset stands out due to its temporal depth, spatial diversity, and multimodal (multiview) perspectives. Each plant is placed on a motorized rotator, enabling image capture from multiple viewing angles at various vertical levels, repeated over multiple days. Such detailed documentation allows researchers to analyze morphological changes, growth rates, and structural variations across different crops and time scales.

2.1.1 Key Characteristics. The following unique characteristics of our dataset make it particularly valuable for vision-based plant modeling, growth analysis, and machine learning research:

- **Multi-Crop and Multi-Instance Diversity:** The dataset includes four agriculturally important crops: wheat, mustard, radish, and okra with multiple plant instances per crop: four for wheat and mustard, five for radish, and two for okra. This diversity promotes the development of crop-agnostic models while capturing intra-class variability in growth patterns and morphology.

- **Growth Cycle Coverage:** Each plant was imaged throughout its life cycle, with the number of observation days varying by crop. Wheat has the longest duration (118 days), followed by okra (86 days), radish (59 days), and mustard (50 days), as illustrated in Figure 2. This temporal density enables detailed time-series analysis like growth analysis, change detection, and phenological stage estimation.
- **Structured Multiview Acquisition:** Imaging was performed at five vertical levels (L1–L5), capturing the plant at different heights to approximate 3D spatial context. At each level, 24 images were taken by rotating the plant in 15° increments (Figure 3), resulting in 120 images per plant per day. This enables learning tasks involving multiview consistency, view synthesis, and geometric reasoning.
- **Controlled Imaging Setup:** The data was collected under consistent indoor lighting and background conditions using a calibrated rotator system. This controlled setup minimizes environmental variability and reduces domain shift, making the dataset well-suited for benchmarking and generalization.

2.1.2 Impact and Applications. The GroMo25 dataset bridges the gap between computer vision and agricultural AI by offering a unique combination of multiview, temporal, and multimodal plant imagery, enabling advancements across multiple domains. For computer vision research, it facilitates progress in multiview learning, neural rendering, 3D reconstruction, and viewpoint-invariant

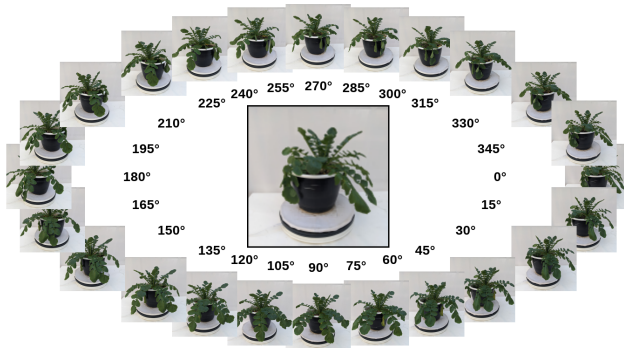


Figure 3: Multi-view imaging of a single potted plant captured at 15° intervals using a rotatory platform. A total of 24 side-view images were taken covering a full 360° rotation, with one additional representative image at the center. This setup enables complete visual coverage of the plant structure.

recognition. Its rich temporal and spatial structure supports self-supervised and contrastive learning. In the context of agricultural AI and plant phenotyping, GroMo25 supports modeling of plant growth over time, detailed analysis of morphological traits (like leaf size and shape), and monitoring of developmental patterns using visual data. Moreover, the dataset serves as a strong foundation for benchmarking tasks focused on temporal consistency and cross-crop generalization. Its structured format makes it useful for tasks like synthetic view generation, domain adaptation, and data augmentation. By enabling realistic reconstruction and simulation of plant growth, GroMo25 can also support the development of interactive virtual reality (VR) environments and digital twins for agriculture. With this versatility, GroMo25 stands as a comprehensive and high-impact resource for advancing vision-based plant monitoring and precision agriculture.

2.2 Tasks and Challenges

The GroMo25 Challenge focuses on advancing plant growth modeling using multiview image data. It comprises two central tasks: plant age prediction and leaf count estimation, both designed to exploit the richness of multiview imaging for a comprehensive understanding of plant development. Each plant is imaged from up to 24 different viewpoints, captured across 5 vertical heights. This spatial diversity provides participants with a detailed and dynamic visual profile of plant structure, offering unique opportunities to explore how appearance varies with view and stage of growth.

2.2.1 Task 1: Plant Age Prediction. The first task involves predicting the chronological age of a plant (in days) based on multiview imagery. Accurate age prediction is fundamental in growth modeling and phenotyping, particularly for understanding growth rates and detecting anomalies in development. Since plants evolve morphologically over time, combining side views helps capture both fine-grained and structural growth features. For instance, Figure 5 illustrates two distinct views of a mustard plant captured on the same day. While the left view shows a more scattered arrangement with a greater number of visible leaves, the right view emphasizes

larger and more mature foliage. Despite being from the same time point, the two views reveal different aspects of the plant’s morphology. This highlights the importance of incorporating multiview information for age prediction, as relying on a single viewpoint may miss crucial growth cues such as leaf size, spread, or density that vary with perspective.

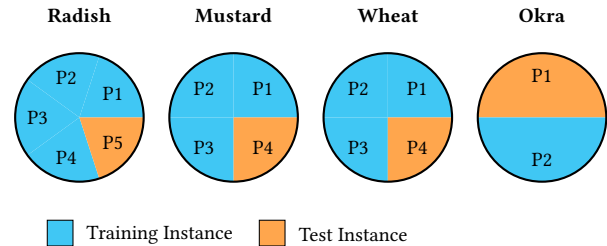


Figure 4: Circular representation of data partitioning for each crop. Each sector (P1–Pn) represents a plant instance. Blue sectors denote training instances, while the orange sector indicates the held-out test instance. For Okra, P1 is used as the test plant and P2 for training.

2.2.2 Task 2: Leaf Count Estimation. The second task focuses on estimating the number of leaves on a plant using multiview images. Leaf counting is a critical proxy for measuring plant health and developmental stage. However, occlusions and leaf overlaps present significant challenges, making single-view estimation unreliable. Figure 6 illustrates the need for multiple views in leaf estimation. In this example, white-labeled leaves are visible from both perspectives, whereas yellow labels indicate leaves that are exclusive to the left and right views, respectively. This shows that a single viewpoint can miss a substantial portion of the plant’s foliage, potentially leading to underestimation. Incorporating multiview data ensures better coverage and enables models to reason about occluded structures.

2.3 Data Partitioning

To ensure a robust evaluation and prevent overfitting to specific plant instances, we adopt a leave-one-out strategy for each crop. As illustrated in Figure 4, each crop is represented by a circular diagram where each sector (P1–Pn) corresponds to a unique plant instance. The blue sectors denote training instances, while the orange sector marks the held-out test instance. For example, in the case of radish, out of five instances, P5 is selected as the test plant, while the rest are used for training and validation. This partitioning ensures that the model is evaluated on its ability to generalize to entirely unseen plants of the same crop, thereby providing a realistic test of generalization performance for age and leaf count prediction tasks.

2.4 Evaluation Metrics

Age prediction can be formulated either as a classification problem (predicting discrete growth stages) or as a regression problem (predicting exact age in days). Similarly, leaf count estimation can be tackled using classification, regression, or instance segmentation approaches. Given the continuous nature of both age and leaf



Figure 5: Left and right views of the same radish plant, captured on the same day, reveal complementary traits: smaller, denser leaves on the left and fewer, larger leaves on the right. This contrast highlights the value of multiview imaging for accurate age prediction.

count, the most appropriate metric for evaluation is the Mean Absolute Error (MAE), which quantifies the average absolute difference between predicted and ground truth values:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (1)$$

where y_i is the ground truth, $\hat{y}_i$ is the predicted value, and N is the total number of samples.

For both tasks, the average MAE is computed across all four crops, ensuring a consistent evaluation metric. This cross-crop aggregation allows a fair comparison across methods while accounting for inter-crop variability in leaf morphology and growth patterns.

Leaderboard and Winner Selection: Two separate leaderboards are prepared, one for each task. Participants are ranked based on their average MAE across all crops. The final winner is selected based on their top performance in both tasks. In our case, the same participant secured the best results for both tasks.

3 Baseline Approach

As a baseline for both tasks in GroMo25, we adopt a Multi-View Vision Transformer (MVVT) [4], inspired by the standard Vision Transformer (ViT) [15], and extend it to process multi-view image inputs. The MVVT model takes a sequence of images captured from 24 viewpoints and jointly encodes spatial and inter-view relationships through a multi-stage pipeline. Each image is first divided into non-overlapping patches via a convolutional patch embedding layer. Positional encodings are then added to preserve spatial structure. To capture dependencies across views, a Multi-view Attention Block (MAB) is added that extends standard self-attention by enabling cross-image interactions. The encoded representations are further refined using a transformer encoder stack. Finally, global mean pooling is applied across all tokens, followed by an MLP head for prediction. This unified architecture supports both regression tasks in GroMo25: leaf count estimation and plant age prediction.

4 Results

In this section, we present the baseline results and the final results achieved by our solution on the GroMo'25 tasks.

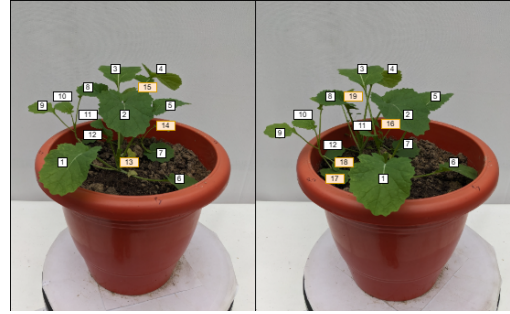


Figure 6: Multi-view annotation of a plant from two angles shows white-labeled leaves visible in both views, while yellow labels mark leaves hidden in the left and right views—underscoring the need for multiview imaging for complete visibility and accurate leaf counting.

4.1 Baseline Results

Table 3 presents the performance of the two tasks: plant age prediction and leaf counting, across four crops: Mustard, Radish, Okra, and Wheat. Evaluation was done using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), where lower values indicate better performance. It can be observed that radish achieved the best results for age prediction (MAE: 5.71), while okra performed best in leaf counting (MAE: 2.04). Mustard and Wheat were the most challenging for age and leaf prediction, respectively.

4.2 Challenge Solution Results

We provide the official leaderboard of the GroMo25 challenge in Table 3, reporting the MAE for both tasks: Age Prediction and Leaf Count Estimation, evaluated across four diverse crops: okra, radish, mustard, and wheat. Each task was assessed independently, and two separate leaderboards are maintained. The final winner is selected based on the best average performance across both tasks.

The top-performing team, **ETTI 2 - VIS**, achieved the lowest average MAE in both tasks, **3.17** for age prediction and **1.99** for leaf count estimation, substantially outperforming all other submissions. Their approach introduced several architectural and training innovations designed specifically for multi-view plant phenotyping. Their approach introduced architectural and training innovations tailored for multi-view plant phenotyping. Built on a ViT backbone (fine-tuned or frozen as needed), they fused multi-view features using a transformer encoder. Their final model takes all 24 views per plant. A key innovation was the use of dynamically weighted selection vectors to prioritize informative views. To enhance rotation invariance, they cyclically permuted input view orders and averaged predictions, improving robustness to camera ordering and view configurations.

Overall, the solution's ability to integrate full-view modeling, transformer-based fusion, dynamic view selection, and ensemble learning contributed to its superior performance, establishing a strong benchmark for future approaches on the GroMo25 dataset.

Table 3: Leaderboard of Mean Absolute Error (MAE) for Age Prediction and Leaf Count tasks across crops and teams.

Rank	Team	Age Prediction MAE				Avg	Leaf Count MAE				Avg
		Okra	Radish	Mustard	Wheat	Age Avg	Okra	Radish	Mustard	Wheat	Leaf Avg
1	ETTI 2 - VIS	1.71	1.61	6.47	2.90	3.17	1.06	0.83	2.70	3.38	1.99
2	DeepLeaf	4.80	4.60	7.80	6.15	5.83	0.99	0.89	7.60	5.25	3.69
3	AlgriTech	3.77	5.03	8.70	8.44	6.48	1.65	1.79	4.40	6.71	3.63
4	PlantPixels	13.10	5.60	3.20	7.30	7.30	2.76	0.66	8.30	5.68	4.35
5	SoumikDas	11.14	2.68	10.18	10.60	8.65	3.06	1.33	3.33	7.20	3.73
6	Resense	12.28	22.6	5.06	7.21	11.78	1.44	4.81	2.16	6.23	3.66
7	Rishi	12.44	10.08	12.64	16.79	12.98	18.50	6.89	4.15	5.18	8.68
8	Agro_Geek	13.42	18.85	11.30	28.45	18.01	1.85	3.98	7.38	30.41	10.90
9	CropIQ	10.80	16.54	21.70	28.60	19.41	2.52	6.48	5.65	8.70	5.83
10	Baseline	5.86	5.71	10.62	8.80	7.74	2.04	4.04	4.99	10.80	5.52

5 Conclusion and Future Directions

We introduced **GroMo25**, the first international challenge on plant growth modeling using multiview imagery, targeting two tasks: age prediction and leaf count estimation across four crops: wheat, mustard, radish, and okra. The challenge attracted participation from multiple teams worldwide and established reliable benchmarks for vision based plant phenotyping. **Team ETTI 2 - VIS** secured the top position in both tasks and was declared the final winner. As part of this challenge, the GroMo25 dataset, comprising high-resolution, temporally and spatially diverse multiview images, will be publicly released to enable further research in computational plant modeling. We believe the dataset will contribute significantly to the development of AI methods in controlled environment agriculture.

The GroMo25 dataset presents opportunities for several exciting directions. Its structured multiview and temporal nature make it suitable for tasks such as self-supervised learning, view synthesis, and simulation-based analysis. It can be used in VR or digital twin platforms to simulate plant growth. It also enables exploration of growth consistency across views and time, aiding the development of general-purpose plant growth models. As a standardized benchmark, GroMo25 supports comparative evaluation and innovation in vision-based phenotyping systems.

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